

Nutrition and Hydration Basics

An overview of macronutrients, micronutrients, food groups, and daily fluid needs

What Is a Balanced Diet?

A balanced diet provides the body with the nutrients it needs in appropriate amounts. It generally includes a variety of foods from different food groups, rather than relying heavily on any single type of food.

Macronutrients

Carbohydrates

Carbohydrates are the body's main source of energy. Whole grains, fruits, vegetables, and legumes provide carbohydrates along with fiber and other nutrients.

Proteins

Proteins support the growth and repair of tissues, including muscles. Sources include lean meat, poultry, fish, eggs, dairy, beans, lentils, nuts, and soy products.

Fats

Fats support cell function and help absorb certain vitamins. Sources such as olive oil, nuts, seeds, and fish provide unsaturated fats, which health bodies generally favor over saturated fats found in some animal and processed foods.

Micronutrients: Vitamins and Minerals

Vitamins and minerals are needed in small amounts but play important roles in processes such as immune function, bone health, and energy metabolism. A varied diet with fruits, vegetables, whole grains, and protein sources typically supplies a wide range of micronutrients.

Food Groups and Portion Basics

Many national dietary guidelines, including those referenced by the NHS Eatwell Guide, suggest a plate made up of:

- A generous portion of vegetables and fruits
- Starchy carbohydrates such as whole grains, potatoes, or rice
- A source of protein such as beans, fish, eggs, or meat
- Some dairy or dairy alternatives
- Small amounts of unsaturated oils or spreads

The WHO recommends limiting free sugars to less than 10% of total daily energy intake, and limiting sodium (salt) intake to less than 2 grams per day for adults.

Understanding Nutrition Labels

Nutrition labels commonly include the following information, which can help with comparing packaged foods:

- Serving size: the amount the listed values refer to
- Calories (energy): the amount of energy provided per serving
- Macronutrient breakdown: grams of carbohydrate, protein, and fat per serving
- Percent Daily Value (%DV): shows how much a nutrient in a serving contributes to a general daily diet, where used

Importance of Hydration

Water supports nearly every function in the body, including regulating temperature, transporting nutrients, and removing waste. The body loses water through sweat, breathing, and urination, and this water needs to be replaced through fluids and food.

How Much Water Do You Need?

Fluid needs vary by individual, activity level, climate, and diet. As a general reference, the U.S. National Academies suggest total daily fluid intakes (from all beverages and food) of around 2.7 liters for adult women and 3.7 liters for adult men, though a meaningful portion of this comes from food. Needs increase with exercise, heat, or illness.

Recognizing Low Fluid Intake

General signs that fluid intake may be low include thirst, dark yellow urine, and tiredness. Drinking water regularly throughout the day, rather than waiting until feeling very thirsty, is a commonly recommended habit.

Healthy Eating Habits

- Plan meals ahead of time to include a variety of food groups
- Eat slowly and pay attention to hunger and fullness cues
- Limit foods and drinks high in added sugar, salt, and saturated fat
- Choose water or unsweetened drinks over sugary beverages where possible
- Read nutrition labels to compare packaged food options

Sources

- World Health Organization (WHO) — Healthy Diet Fact Sheet

- U.S. National Institutes of Health (NIH) — National Academies Dietary Reference Intakes for Water
- UK National Health Service (NHS) — The Eatwell Guide
- U.S. Centers for Disease Control and Prevention (CDC) — Nutrition Basics and Water and Healthier Drinks

This document provides general educational information only. It is not medical advice, diagnosis, or treatment. Consult a qualified healthcare provider for individual health concerns.